While Loop

A while loop is used to execute a block of code repeatedly as long as the given condition is True

In [3]:

```
# print 1 to 5:
i=1
while i<6:
    print(i)
    i=i+1
```

```
1
2
3
4
5
```

Practice Question

1. Write a program to print numbers from 1 to 10 using a while loop.

In [7]:

```
i=1
while i<=10:
    print(i)
    i +=1
```

```
1
2
3
4
5
6
7
8
9
10
```

In [9]:

```
i=1
while i<=10:
    print(i,end=" ")
    i +=1
```

```
1 2 3 4 5 6 7 8 9 10
```

2. Write a program to print numbers from 10 to 1.

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In [1]:

```
num=10
while num >=1:
    print(num)
    num -=1
```

```
10
9
8
7
6
5
4
3
2
1
```

3. Write a program to find the sum of numbers from 1 to N.

In [3]:

```
num=1
sum=0
while num <=5:
    sum = sum +num
    num = num +1
print(sum)
```

```
15
```

4. Write a program to find the factorial of a number.

In [10]:

```
a = 5
f = 1
while a >= 1:
    f = f * a
    a = a - 1
print(f)
```

```
120
```

5. Write a program to reverse a number.

In [13]:

```
a=123
rev=0
while a > 0:
    d=a%10
    rev=rev *10+d
    a //=10
print(rev)
```

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6. Write a program to count the number of digits in a number.

In [3]:

```
a=12345
c=0
while a>0:
    c=c+1
    a=a//10
print(c)
```

5

7. Write a program to find the sum of digits of a number.

In [8]:

```
a=12345
s=0
while a>0:
    d=a % 10
    s=s+d
    a=a//10
print(s)
```

15

8. Write a program to check whether a number is a palindrome.

In [18]:

```
a=121
og=a
rev=0
while a>0:
    d=a%10
    rev=rev * 10+d
    a //=10
if og==rev:
    print("palindrome")
else:
    print("not palindrome")
```

palindrome

9. Write a program to check whether a number is an Armstrong number.

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In [20]:

```
a=157
og=a
rev=0
p=len(str(a))
while a>0:
    d=a %10
    rev = rev+d **p
    a //=10
if og==rev:
    print("Armstrong")
else:
    print("not Armstrong")
```

```
not Armstrong
```

10. Write a program to check whether a number is prime or not.

In [1]:

```
a = 4
i = 2

while i < a:
    if a % i == 0:
        print("Not Prime")
        break
    i = i + 1
else:
    if a > 1:
        print("Prime")
    else:
        print("Not Prime")
```

```
Not Prime
```

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